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Course: 21CSC202J - Operating Systems

Unit – 3 : Session – 14 : SLO – 2 | Resource-Allocation-Graph Algorithm — Deadlock Check

Matrices from the graph

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Instances: r1 = 2, r2 = 3, r3 = 2
Process   Allocation (r1 r2 r3)   Request (r1 r2 r3)
p0        1  0  1                0  1  1
p1        1  1  0                1  0  0
p2        0  1  0                0  0  1
p3        0  1  0                1  2  0
Allocated = (2 3 1), so Available = (2 3 2) - (2 3 1) = (0 0 1)
```

Safety check

p2: request (0 0 1) <= (0 0 1) -> finishes, Available = (0 1 1)

p0: request (0 1 1) <= (0 1 1) -> finishes, Available = (1 1 2)

p1: request (1 0 0) <= (1 1 2) -> finishes, Available = (2 2 2)

p3: request (1 2 0) <= (2 2 2) -> finishes, Available = (2 3 2)

Answer

1. The system is NOT in a deadlock state.
2. Safe sequence: <p2, p0, p1, p3>